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#!/usr/bin/env python3
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"""
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ans_test_runner.py - runs the ANS Forth test suite (ans_tests/) against a  
live forth6809 image, over either a MAME "-bitb" TCP socket or a real  
serial-connected MECB6809 with the Forth binary in EPROM.
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Why this exists (vs. minicom): forth6809's outer interpreter (QUIT) is line-based - it reads one line via ACCEPT, interprets it, then prints "\r\n ok" (or "\r\n ERROR n" on a thrown exception) before reading the next line. That "ok"/"ERROR" reply is a natural, content-based ready signal. Instead of pasting a whole file at a fixed baud rate and hoping end-of-character delays are long enough (the approach currently being fought with minicom, and the subject of the ongoing interrupt/RTS handshaking investigation), this script sends exactly one line, then BLOCKS until it sees that line's own "ok"/"ERROR" reply before sending the next one. That makes the pacing self-adjusting to however fast the target actually is - MAME under the debugger, MAME free-running, or real hardware - with no delay constant to tune per platform.

Three things this script does, per the project's outstanding ANS-test task list:

1. Loads the [IF]/[ELSE]/[THEN] tool-extension words first (00a_tool_ext_conditionals.fs) - forth6809.asm doesn't implement them, but ttester.fs needs them immediately (its own HAS-FLOATING / HAS-FLOATING-STACK checks use [IF]/[ELSE]/[THEN]).
2. Loads the shared preamble: ttester.fs, then 00_test_prelude.fs.
3. Loads one or more per-section *.tests.fs files. Each of those files is now self-contained: 09_outer_interpreter.tests.fs carries its own copy of GT1/GT2 rather than depending on section 13 having run first, and every section file opens with "MARKER Mnn" and closes by invoking Mnn, which erases everything that file itself defined (confirmed against forth6809.asm's MARKERW/DOMARKER - it snapshots and restores DPHERE/CODEHERE/VARHERE/LATEST, i.e. dictionary, value-space and the search-chain all at once). That makes each section file independently runnable, in any order, any number of times, from a single shared harness+prelude load.

Long test-file lines are re-wrapped before sending: forth6809's TIB is 80 bytes (TIBBUFL EQU 80 in forth6809.asm), but several *.tests.fs lines run past 1000 characters (many "T{ ... }T" blocks concatenated on one source line, harmless to a file-based INCLUDE but fatal to a terminal that ACCEPTs 80 bytes per line). Re-wrapping only ever splits at whitespace outside of S"/."-delimited strings, so it never changes the

token stream Forth sees, and never splits a "\"" line comment (which would turn its tail into live code on the next line).

Usage examples:

```
# Against MAME (spawns it, using the same -rs232 null_modem -bitb
# socket approach as run_all_tests.sh):
python3 ans_test_runner.py --target mame \
    --sections 08 09 10 11 12 13 14 15 16 17 18 19 20 21 23 24 26

# Against real hardware over a serial port:
python3 ans_test_runner.py --target serial --serial-port /dev/ttyUSB0 \
    --baud 19200 --sections 15
```

Run with --help for every option and its default.

"""

```
import argparse
import re
import socket
import subprocess
import sys
import time
from pathlib import Path
```

```
# -----
# Constants matching forth6809.asm
# -----
TIBBUFL = 80          # forth6809.asm: TIBBUFL EQU 80
SAFE_LINE_WIDTH = 72 # leave headroom below TIBBUFL for CR and slop
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OK_MARKER = b" ok"
ERROR_MARKER = b" ERROR "
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# ttester.fs's own failure messages (from ERROR1's S" ... " text, per
# ttester.fs read directly out of this project's ans_tests/ directory)
# plus forth6809's own system-level abort message. Any of these
# appearing in a section's captured transcript means that section did
# NOT fully pass, regardless of whether "ok" also appeared.
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FAILURE_PATTERNS = [
    b"INCORRECT RESULT:",
    b"WRONG NUMBER OF RESULTS:",
    b"INCORRECT CELL RESULT:",
    b"NUMBER OF CELL RESULTS BEFORE '->' DOES NOT MATCH",
    b"NUMBER OF CELL RESULTS BEFORE AND AFTER '->' DOES NOT MATCH:",
    b"NUMBER OF RESULTS AFTER '->' BELOW ...}T SPECIFICATION:",
    b"INCORRECT FP RESULT:",
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        b"WRONG NUMBER OF FP RESULTS:",
        ERROR_MARKER,
    ]

DEFAULT_SECTIONS = [
    "08", "09", "10", "11", "12", "13", "14", "15", "16",
    "17", "18", "19", "20", "21", "23", "24", "26",
]

# -----
# Line re-wrapping: split long lines only on whitespace outside
# S"/."-delimited strings, and never touch a "\" comment line.
# -----
def is_line_comment(stripped):
    leading = stripped.lstrip(" \t")
    return leading.startswith("\\") and (len(leading) == 1 or leading[1] in " \t")

def tokenize_preserving_strings(stripped):
    """Split on whitespace, but keep an S"/./C"/ABORT"-introduced string
    literal as one atomic token so a caller never mistakes a character
    inside it (a stray ':' or ';' in test text, say) for a real word.
    Shared by rewrap_line (so a rewrap point never lands inside a
    string) and colon_delta (so a ':' or ';' inside a string is never
    mistaken for a real colon-definition boundary)."""
    tokens = []
    i = 0
    n = len(stripped)
    while i < n:
        while i < n and stripped[i] in " \t":
            i += 1
        if i >= n:
            break
        start = i
        # Detect a string-introducing word: S" / ." / C" / ABORT"
        # followed by exactly one space then a '"'-terminated string.
        word_end = i
        while word_end < n and stripped[word_end] not in " \t":
            word_end += 1
        word = stripped[i:word_end]
        if word in ('S"', '."', 'C"', 'ABORT"') and word_end < n and stripped[word_end] == ' ':
            close = stripped.find('"', word_end + 1)
            if close != -1:
                tokens.append(stripped[start:close + 1])
                i = close + 1
                continue

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        tokens.append(stripped[start:word_end])
        i = word_end
    return tokens

def colon_delta(line):
    """Net change in open-colon-definition depth this line causes,
    scanning tokens outside string literals via tokenize_preserving_strings
    so a ':' or ';' inside test text is never mistaken for a real one.
    ':' opens one level, ';' closes one - matched as exact tokens, not
    substrings, so words like ':' inside a larger token don't count."""
    stripped = line.strip("\r\n").strip()
    if stripped == "" or is_line_comment(stripped):
        return 0
    delta = 0
    for tok in tokenize_preserving_strings(stripped):
        if tok == ":":
            delta += 1
        elif tok == ";":
            delta -= 1
    return delta

def rewrap_line(line, width=SAFE_LINE_WIDTH):
    stripped = line.strip("\r\n")
    if len(stripped) <= width:
        return [stripped] if stripped != "" or True else []

    if is_line_comment(stripped.lstrip(" \t")):
        # A "\" line comment: never split (its tail would become live
        # code on a continuation line). Just send it long - the target
        # discards it as a comment before hitting any length limit
        # inside its own text-parsing (parses one BL-delimited "word"
        # per token normally, but the whole rest of the line is inside
        # a comment, so nothing is ever an over-length token here).
        return [stripped]

    tokens = tokenize_preserving_strings(stripped)

    lines = []
    cur = ""
    for tok in tokens:
        candidate = tok if cur == "" else cur + " " + tok
        if len(candidate) > width and cur != "":
            lines.append(cur)
            cur = tok
        else:

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        cur = candidate
    if cur:
        lines.append(cur)
    return lines

def rewrap_source(text):
    out = []
    for line in text.split("\n"):
        out.extend(rewrite_line(line))
    return out

# -----
# Transports
# -----
class SocketTransport:
    """Binds and listens BEFORE the caller spawns MAME, exactly like
    mame_listener.py does - MAME's own "-bitb socket.HOST:PORT" tries to
    CONNECT as a client first, so it connects straight in with no device
    path to discover and no startup race."""

    def __init__(self, host, port, accept_timeout):
        self.srv = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        self.srv.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
        self.srv.bind((host, port))
        self.srv.listen(1)
        self.srv.settimeout(accept_timeout)
        self.conn = None

    def accept(self):
        self.conn, addr = self.srv.accept()
        self.conn.settimeout(0.2)
        return addr

    def write(self, data):
        self.conn.sendall(data)

    def read_some(self):
        try:
            data = self.conn.recv(4096)
        except socket.timeout:
            return b""
        if data == b"":
            raise ConnectionResetError("connection closed by peer")
        return data

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def close(self):
    try:
        if self.conn:
            self.conn.close()
    finally:
        self.srv.close()

class SerialTransport:
    """Real serial-connected MECB6809, Forth binary in EPROM. Requires
    pyserial (pip install pyserial). rtscts=True enables hardware flow
    control at the host-OS/driver level, independent of whatever
    forth6809's own RTS/CTS firmware logic is doing - it is the
    project's stated eventual target alongside MAME."""

    def __init__(self, port, baud, rtscts):
        import serial # local import: only required for this transport
        self.ser = serial.Serial(
            port=port, baudrate=baud, timeout=0.2, rtscts=rtscts,
        )

    def write(self, data):
        self.ser.write(data)
        self.ser.flush()

    def read_some(self):
        return self.ser.read(4096)

    def close(self):
        self.ser.close()

# -----
# MAME process management (mirrors run_all_tests.sh's approach)
# -----
def launch_mame(args):
    cmd = [
        args.mame_bin, args.mame_system,
        "-rs232", "null_modem",
        "-bitb", f"socket.{args.listener_host}:{args.listener_port}",
        "-debug", "-debugscript", args.retrigger_cmd,
        "-window", "-resolution", "640x480",
    ]
    print(f"[mame] launching: {' '.join(cmd)}", file=sys.stderr)
    return subprocess.Popen(cmd, stdout=subprocess.DEVNULL, stderr=subprocess.DEV

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def assemble_rom(args):
    cmd = [
        args.lwasmbin, "--6809", "--format=raw",
        f"--output={args.rom_out}", f"--list={args.rom_out}.lst",
        f"--define=UNITTESTS={args.unittests}",
        f"--define=SERIALPOLL={args.serialpoll}",
        args.asm_source,
    ]
    print(f"[asm] {' '.join(cmd)}", file=sys.stderr)
    subprocess.run(cmd, check=True)
    Path(args.rom_out).replace(args.rom_dest)

# -----
# The line-at-a-time send/wait protocol
# -----

class LineResult:
    def __init__(self, sent, echoed, ok, error, timed_out):
        self.sent = sent
        self.echoed = echoed
        self.ok = ok
        self.error = error
        self.timed_out = timed_out

def send_line_and_wait(transport, line, char_delay, reply_timeout, log):
    """Send one line, terminated by CR (forth6809's ACCEPT/QUIT treat CR
    ($0D) as end-of-line and silently ignore LF - see ACCEPT's ALOOP:
    CMPB #13 -> ADONE, CMPB #10 -> BRA ALOOP - so CR alone is correct and
    sending CRLF would just cost one ignored byte per line for no
    benefit)."""
    payload = line.encode("ascii", errors="replace") + b"\r"
    if char_delay > 0:
        for b in payload:
            transport.write(bytes([b]))
            time.sleep(char_delay)
    else:
        transport.write(payload)

    buf = b""
    deadline = time.monotonic() + reply_timeout
    saw_ok = False
    saw_error = False
    while time.monotonic() < deadline:
        chunk = transport.read_some()
        if chunk:
            buf += chunk

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log.write(chunk)
log.flush()
if OK_MARKER in buf:
    saw_ok = True
    break
if ERROR_MARKER in buf:
    saw_error = True
    # keep draining briefly to catch the throw-code digits
    drain_deadline = time.monotonic() + 0.5
    while time.monotonic() < drain_deadline:
        extra = transport.read_some()
        if extra:
            buf += extra
            log.write(extra)
            drain_deadline = time.monotonic() + 0.5
    break
timed_out = not (saw_ok or saw_error)
return LineResult(payload, buf, saw_ok, saw_error, timed_out), buf

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def send_file(transport, path, char_delay, reply_timeout, retries, log, transcrip
    soft_timeout=False, grace_timeout=1.0):
    """Two distinct, known cases produce fewer "ok" replies than lines
    sent, and both are correct target behavior, not data loss or a
    transport bug - so neither should cost a full reply_timeout stall
    per line, and neither should be reported as a [FAIL]:

```

1. Mid colon-definition (tracked here via colon_delta, purely from the source text - no target state needed). QOK only prints "ok" once STATE is back to 0, i.e. once the definition's ":" has been matched by a ";" - every interior line of a multi-line ": ... ;" gets no "ok" at all, in any file, not just the harness ones. A real compile-time error still prints immediately regardless of STATE, so a short grace_timeout (rather than 0) still catches that quickly without paying the full reply_timeout on every interior line of every multi-line definition.
2. soft_timeout=True, for harness files with multi-line [IF]/[ELSE] blocks (ttester.fs's own HAS-FLOATING/HAS-FLOATING-STACK checks). [ELSE] (00a_tool_ext_conditionals.fs, the ANS standard's own reference implementation) skips a false branch by calling REFILL directly from inside its own BL WORD loop whenever it runs out of the current line - entirely inside the single INTERPRET call QLOOP is already waiting on. Every line [ELSE] skips past is read by REFILL, not a fresh QUERY/ACCEPT cycle, and QOK doesn't reply until the whole multi-line skip finally returns - one reply for the entire span, not one per physical line. This can't be predicted from the source alone (it depends on ENVIRONMENT?'s

runtime answer), so it still costs a wait, but grace_timeout keeps that wait short instead of the full reply_timeout.

```
In both cases attempt=1 is final - retrying is unsafe here regardless
(see the mid-definition retry note in the module docstring history:
resending would re-feed the same text into a still-open definition,
or into REFILL's own skip, a second time)."""
text = Path(path).read_text()
lines = rewrap_source(text)
fail_hit = False
depth = 0
for line in lines:
    if line.strip() == "":
        continue
    depth = max(0, depth + colon_delta(line))
    expect_ok = (depth == 0)
    line_timeout = reply_timeout if expect_ok else min(reply_timeout, grace_t

attempt = 0
while True:
    attempt += 1
    result, buf = send_line_and_wait(transport, line, char_delay, line_ti
    transcript.extend(buf.split(b"\n"))

    if not expect_ok:
        if result.error:
            # A real compile-time error mid-definition - report
            # it, but don't retry (unsafe mid-definition) and
            # don't wait further; classify()'s transcript-wide
            # scan for " ERROR " still catches it either way.
            print(f"      [ERROR mid-definition] {line[:60]!r}", file=sys.s
            fail_hit = True
            break

        if result.timed_out and soft_timeout:
            print(f"      [note] no immediate reply for: {line[:60]!r} "
                  f"(expected while inside a multi-line [IF]/[ELSE] skip)",
                  file=sys.stderr)
            break

        if result.timed_out and attempt <= retries:
            print(f"      [retry {attempt}] no ok/ERROR within {line_timeout}s
                  file=sys.stderr)
            continue

        if result.timed_out:
            print(f"      [FAIL] gave up after {retries} retries on: {line[:60]
            fail_hit = True
            break
```

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    return fail_hit

def classify(transcript_bytes):
    for pattern in FAILURE_PATTERNS:
        if pattern in transcript_bytes:
            return False
    return True

def run_section(transport, ans_dir, section, char_delay, reply_timeout, retries,
    matches = sorted(Path(ans_dir).glob(f"{section}_*.tests.fs"))
    if not matches:
        print(f"[skip] no test file found for section {section}", file=sys.stderr)
        return None
    path = matches[0]
    print(f"=== section {section} ({path.name}) ===")
    log_path = Path(log_dir) / f"{section}.raw.log"
    transcript = []
    with open(log_path, "wb") as log:
        timed_out_any = send_file(transport, path, char_delay, reply_timeout, ret
                                grace_timeout=grace_timeout)
    transcript_bytes = b"\n".join(transcript)
    passed = classify(transcript_bytes) and not timed_out_any
    print(f"    -> {'PASS' if passed else 'FAIL'} (log: {log_path})")
    return passed

def main():
    p = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.Raw
    p.add_argument("--target", choices=["name", "serial"], required=True)
    p.add_argument("--ans-dir", default="ans_tests", help="directory containing a li
    p.add_argument("--sections", nargs="+", default=DEFAULT_SECTIONS,
        help="section number prefixes to run, e.g. 08 09 10 (default:
    p.add_argument("--char-delay", type=float, default=0.0,
        help="extra seconds between individual characters within a li
    p.add_argument("--reply-timeout", type=float, default=10.0,
        help="seconds to wait for 'ok'/'ERROR' after a line before tr
    p.add_argument("--retries", type=int, default=2,
        help="retries per line before giving up (default: 2)")
    p.add_argument("--grace-timeout", type=float, default=1.0,
        help="seconds to wait for a reply on a line that is known in
            "(interior lines of a multi-line colon-definition; defau
            "since it only needs to catch a genuine compile-time err
    p.add_argument("--log-dir", default="ans_test_results")

    # name-target options (mirror run_all_tests.sh's flags/defaults)

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p.add_argument("--asm-source", default="forth6809.asm")
p.add_argument("--lwasm-bin", default="lwasm")
p.add_argument("--mame-bin", default="./mecb6809")
p.add_argument("--mame-system", default="mecb6809")
p.add_argument("--rom-out", default="forth6809.bin")
p.add_argument("--rom-dest", default=str(Path.home() / "Library/Application S
p.add_argument("--retrigger-cmd", default="retrigger.cmd")
p.add_argument("--listener-host", default="127.0.0.1")
p.add_argument("--listener-port", type=int, default=2000)
p.add_argument("--serialpoll", type=int, default=1, choices=[0, 1],
                help="assemble with polling (1, default - avoids the interrup
                    "under separate investigation) or interrupt-driven (0) s
p.add_argument("--unittests", type=int, default=0, choices=[0, 1],
                help="0 (default): a normal end-user image, which is what the
p.add_argument("--skip-assemble", action="store_true",
                help="reuse whatever ROM is already at --rom-dest instead of
p.add_argument("--mame-accept-timeout", type=float, default=30.0,
                help="seconds to wait for MAME's initial connection (default:

# serial-target options
p.add_argument("--serial-port")
p.add_argument("--baud", type=int, default=19200)
p.add_argument("--rtscts", action="store_true", help="enable hardware RTS/CTS

args = p.parse_args()

Path(args.log_dir).mkdir(parents=True, exist_ok=True)

mame_proc = None
if args.target == "mame":
    if not args.skip_assemble:
        assemble_rom(args)
    transport = SocketTransport(args.listener_host, args.listener_port, args.
    mame_proc = launch_mame(args)
    try:
        addr = transport.accept()
        print(f"[mame] connected from {addr}", file=sys.stderr)
    except socket.timeout:
        print("FAIL: MAME never connected - check -rs232/-bitb support with -
            mame_proc.terminate()
            sys.exit(2)
    else:
        if not args.serial_port:
            p.error("--serial-port is required for --target serial")
        transport = SerialTransport(args.serial_port, args.baud, args.rtscts)

try:

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# Harness + preamble, loaded once, ahead of every section.
harness_log = Path(args.log_dir) / "00_harness.raw.log"
transcript = []
with open(harness_log, "wb") as log:
    for fname in ("00a_tool_ext_conditionals.fs", "ttester.fs", "00_test_
        fpath = Path(args.ans_dir) / fname
        print(f"=== loading {fname} ===")
        send_file(transport, fpath, args.char_delay, args.reply_timeout,
            soft_timeout=True, grace_timeout=args.grace_timeout)

results = {}
for section in args.sections:
    results[section] = run_section(
        transport, args.ans_dir, section,
        args.char_delay, args.reply_timeout, args.retries, args.log_dir,
        grace_timeout=args.grace_timeout,
    )

print("\n=== summary ===")
n_pass = sum(1 for v in results.values() if v)
n_fail = sum(1 for v in results.values() if v is False)
n_skip = sum(1 for v in results.values() if v is None)
for section, v in results.items():
    label = "PASS" if v else ("SKIP" if v is None else "FAIL")
    print(f" {section}: {label}")
print(f"{n_pass} passed, {n_fail} failed, {n_skip} skipped")
sys.exit(0 if n_fail == 0 else 1)

finally:
    transport.close()
    if mame_proc is not None:
        mame_proc.terminate()
        try:
            mame_proc.wait(timeout=5)
        except subprocess.TimeoutExpired:
            mame_proc.kill()

if __name__ == "__main__":
    main()

```